Rydberg Atomic Receiver Simulation
Complete Numerical Parameter Sheet

Parameters used for Fig. 3–Fig. 7 simulations
Paper values, physical constants, derived values, and implementation-specific quantities

How to Use This Sheet

This document is intended as a quick-reference sheet for the M.Tech. thesis simulation.

Three categories are used throughout:

- **PAPER-STATED:** directly reported in the reference paper or supplementary material.
- **DERIVED:** calculated directly from paper-stated quantities using the stated equations.
- **IMPLEMENTATION / CALIBRATION:** a numerical choice made in the present reproduction code because the paper does not completely specify the corresponding numerical procedure.
- **UNRESOLVED:** a quantity for which the paper does not provide enough information for a unique reconstruction.

Important: values labelled implementation/calibration must not be presented to the guide as if they were explicitly printed in the paper.

1 1. Physical Constants

Table 1: Physical constants used in the numerical implementation.

Quantity	Symbol	Value	Unit
Boltzmann constant	k_B	1.380649×10^{-23}	J/K
Reduced Planck constant	$\hbar$	$1.054571817 \times 10^{-34}$	J s
Elementary charge	e	$1.602176634 \times 10^{-19}$	C
Bohr radius	a_0	$5.29177210903 \times 10^{-11}$	m
Vacuum permittivity	ϵ_0	8.854×10^{-12}	F/m
Free-space impedance	η_0	377	Ω
Speed of light	c	2.99792458×10^{8}	m/s

Note: the frozen Fig. 3 implementation currently retains rounded values

$$e = 1.6 \times 10^{-19}\ \text{C}, \qquad a_0 = 5.2 \times 10^{-11}\ \text{m},$$

whereas the reference parameter sheet gives the standard precise values above. These should not be mixed silently between simulations.

2 2. Atomic System

The receiver is modelled as a four-level cesium system,

$$|1\rangle \to |2\rangle \to |3\rangle \to |4\rangle,$$

corresponding to

Table 2: Atomic-system parameters.

Parameter	Symbol	Value	Unit
Vapor-cell length	L	1 cm	m
Atomic density	N_0	1×10^{15}	m^{-3}
State $\|1\rangle$ decay	γ_1	0	rad/s
State $\|2\rangle$ decay	$\gamma_2/(2\pi)$	5.2	MHz
State $\|3\rangle$ decay	$\gamma_3/(2\pi)$	3.9	kHz
State $\|4\rangle$ decay	$\gamma_4/(2\pi)$	1.7	kHz
RF transition dipole	$\wp_{\mathrm{RF}}$	$-1443.459\, ea_0$	C m
Probe transition dipole	$\wp_{12}$	$2.5\, ea_0$	C m
RF carrier frequency	f_{RF}	3.5	GHz
Probe wavelength	λ_p	852	nm
Coupling wavelength	λ_c	510	nm
Probe Rabi frequency	$\Omega_p/(2\pi)$	8	MHz
Coupling Rabi frequency	$\Omega_c/(2\pi)$	1	MHz

$$6S_{1/2} \to 6P_{3/2} \to 47D_{5/2} \to 48P_{3/2}.$$

For the numerical implementation,

$$\gamma_2 = 2\pi(5.2 \times 10^6), \tag{1}$$
$$\gamma_3 = 2\pi(3.9 \times 10^3), \tag{2}$$
$$\gamma_4 = 2\pi(1.7 \times 10^3), \tag{3}$$
$$\Omega_p = 2\pi(8 \times 10^6), \tag{4}$$
$$\Omega_c = 2\pi(1 \times 10^6). \tag{5}$$

The RF transition dipole is

$$\wp_{\mathrm{RF}} = -1443.459\, e\, a_0. \tag{6}$$

The probe dipole quantity used by the frozen Fig. 3 code is

$$\wp_{12} = (2.5ea_0)^2. \tag{7}$$

3 3. Probe Optical Parameters

Table 3: Probe laser and optical parameters.

Quantity	Symbol	Value	Unit
Probe wavelength	λ_p	852	nm
Coupling wavelength	λ_c	510	nm
Probe Rabi frequency	$\Omega_p/(2\pi)$	8	MHz
Coupling Rabi frequency	$\Omega_c/(2\pi)$	1	MHz
Probe beam diameter	d	0.76	mm
Probe beam radius used by frozen code	d_{code}	0.38	mm
Coupling beam diameter	–	1.95	mm
Cell length	L	1	cm

The paper explicitly calls 0.76 mm the $1/e^2$ beam diameter.
However, the frozen Fig. 3 code uses

$$d_{\text{code}} = 0.38 \text{ mm} \tag{8}$$

as an empirical calibration choice.
This is **not** a paper-stated radius interpretation.

4 4. Probe Input Power

The paper's Eq. (3) is

$$P_{\text{in}} = \frac{\pi}{2\eta_0}\left(\frac{d\Omega_p\hbar}{2\sqrt{\wp_{12}}}\right)^2. \tag{9}$$

Using the frozen Fig. 3 calibration

$$d = 0.38 \text{ mm},$$

gives

$$\boxed{P_{\text{in}} = 9.7690\ \mu\text{W}.} \tag{10}$$

For comparison, using the paper's literal stated diameter,

$$d = 0.76 \text{ mm},$$

gives

$$P_{\text{in}} = 39.0761\ \mu\text{W}. \tag{11}$$

Therefore,

$$\frac{39.0761}{9.7690} \approx 4. \tag{12}$$

The factor of four follows directly from the squared dependence on d.

5 5. Susceptibility and Optical Transmission

The susceptibility coefficient is

$$C_0 = -\frac{2N_0\wp_{12}}{\epsilon_0\hbar\Omega_p}. \tag{13}$$

Using the rounded constants in the frozen Fig. 3 implementation,

$$\boxed{C_0 \approx -1.84362 \times 10^{-5}.} \tag{14}$$

The susceptibility is

$$\chi(t) = C_0\rho_{21}(t). \tag{15}$$

The probe output power is

$$\boxed{P_{\text{out}}(t) = P_{\text{in}} \exp[-k_p L \ \text{Im}\{\chi(t)\}]} \tag{16}$$

where

$$k_p = \frac{2\pi}{\lambda_p}. \tag{17}$$

For $\lambda_p = 852$ nm,

$$k_p \approx 7.37463 \times 10^6 \text{ rad/m}. \tag{18}$$

No additional plotting normalization or scaling factor is used.

6 6. Four-Level Hamiltonian

All master-equation frequencies are angular frequencies in rad/s.

The Hamiltonian used in the numerical model is

$$
\begin{aligned}
H = & -\Delta_p|2\rangle\langle 2| - (\Delta_p + \Delta_c)|3\rangle\langle 3| && (19)\\
& - (\Delta_p + \Delta_c + \Delta_{\mathrm{RF}})|4\rangle\langle 4| && (20)\\
& + \frac{\Omega_p}{2}\left(|1\rangle\langle 2| + |2\rangle\langle 1|\right) && (21)\\
& + \frac{\Omega_c}{2}\left(|2\rangle\langle 3| + |3\rangle\langle 2|\right) && (22)\\
& + \frac{\Omega_{\mathrm{RF}}}{2}\left(|3\rangle\langle 4| + |4\rangle\langle 3|\right). && (23)
\end{aligned}
$$

The steady-state density matrix is obtained from

$$\rho_{\mathrm{ss}} = \mathrm{steadystate}(H, \{C_i\}). \tag{24}$$

The optical response is obtained from

$$\rho_{21} = \langle 2|\rho_{\mathrm{ss}}|1\rangle. \tag{25}$$

7 7. Decay / Collapse Operators

The cascade decay model uses

$$
\begin{aligned}
C_2 &= \sqrt{\gamma_2}|1\rangle\langle 2|, && (26)\\
C_3 &= \sqrt{\gamma_3}|2\rangle\langle 3|, && (27)\\
C_4 &= \sqrt{\gamma_4}|3\rangle\langle 4|. && (28)
\end{aligned}
$$

Thus,

$$
\begin{aligned}
\gamma_2/(2\pi) &= 5.2\ \mathrm{MHz}, && (29)\\
\gamma_3/(2\pi) &= 3.9\ \mathrm{kHz}, && (30)\\
\gamma_4/(2\pi) &= 1.7\ \mathrm{kHz}. && (31)
\end{aligned}
$$

8 8. LO-Free Receiver Parameters

Table 4: LO-free receiver parameters.

Quantity	Symbol	Value	Unit
RF Rabi frequency	$\Omega_{\mathrm{RF}}/(2\pi)$	6	MHz
Observation uncertainty	$\tilde{\epsilon}$	0.5%	–
QPN variance	σ^2_{QPN}	0	implementation
AT threshold	$\Omega_{\mathrm{RF,th}}/(2\pi)$	5.500	MHz

The LO-free noise variance is

$$\sigma^2_{\mathrm{Ry}} = \sigma^2_{\mathrm{BGN}} + \sigma^2_{\mathrm{QPN}} + \tilde{\epsilon}^2 P_{\mathrm{Rx}}. \tag{32}$$

The corresponding SNR is

$$\boxed{\mathrm{SNR}_{\mathrm{Ry}} = \frac{P_{\mathrm{Rx}}|h|^2}{\sigma^2_{\mathrm{BGN}} + \sigma^2_{\mathrm{QPN}} + \tilde{\epsilon}^2 P_{\mathrm{Rx}}}}. \tag{33}$$

The current implementation uses

$$\sigma^2_{\mathrm{QPN}} = 0 \tag{34}$$

as a paper-licensed simplification.

9 9. LO-Dressed Receiver Parameters

Table 5: LO-dressed receiver parameters.

Quantity	Symbol	Value	Unit
LO Rabi frequency	$\Omega_{\mathrm{LO}}/(2\pi)$	4.23	MHz
RF Rabi frequency	$\Omega_{\mathrm{RF}}/(2\pi)$	20	kHz
RF–LO frequency difference	Δ_f	150	kHz
Coupling detuning	Δ_c	0	MHz
LNA gain	G_{LNA}	20	dB
LNA gain, linear	G_{LNA}	100	–
Load resistance	R_L	50	Ω
Photodiode responsivity	D	0.55	A/W
Temperature	T	290	K
Photon efficiency	η_{eff}	0.8	–
Phase used in current implementation	$\Delta\phi$	0	rad

The phase

$$\Delta\phi = 0 \tag{35}$$

is an **implementation assumption**, not a paper-stated parameter.

10 10. Wireless / Communication Parameters

Table 6: Wireless communication parameters.

Quantity	Symbol	Value	Unit
Transmit antenna gain	G_{Tx}	2.15	dBi
Transmit gain, linear	G_{Tx}	1.640590	–
Receive antenna gain	G_{Rx}	2.15	dBi
Receive gain, linear	G_{Rx}	1.640590	–
RF carrier frequency	f_{RF}	3.5	GHz
RF wavelength	λ_{RF}	0.0856550	m
Background noise power	σ^2_{BGN}	−90 dBm	W-equivalent
Background noise in watts	σ^2_{BGN}	1.0×10^{-12}	W
Bandwidth	B	1	MHz
Temperature	T	290	K

The dBm conversion is

$$P_{\mathrm{W}} = 10^{(P_{\mathrm{dBm}}-30)/10}. \tag{36}$$

The gain conversion is

$$G_{\text{linear}} = 10^{G_{\text{dB}}/10}. \tag{37}$$

11 11. Received RF Power and RF Rabi Frequency

The Rydberg receiver uses

$$P_{\text{Rx}} = \frac{P_{\text{Tx}} G_{\text{Tx}} \eta_0}{4\pi d^2}. \tag{38}$$

The RF field magnitude is related to the received quantity through

$$|E_{\text{RF}}| = \sqrt{P_{\text{Rx}}}, \tag{39}$$

under the paper's adopted convention.
The RF Rabi frequency is

$$\boxed{\Omega_{\text{RF}} = \frac{|\wp_{\text{RF}}|}{\hbar} \sqrt{P_{\text{Rx}}}.} \tag{40}$$

12 12. Conventional Receiver

The effective received power used in the present implementation is

$$P_c(d) = \left(\frac{P_{\text{Tx}} G_{\text{Tx}}}{4\pi d^2} \right) \left(\frac{\lambda_{\text{RF}}^2}{4\pi} \right). \tag{41}$$

The conventional noise variance is

$$\boxed{\sigma_{\text{Conv}}^2 = G_{\text{Rx}} G_{\text{LNA}} \sigma_{\text{BGN}}^2 + 4 k_B T B.} \tag{42}$$

For the current parameter set,

$$4 k_B T B = 1.601553 \times 10^{-14} \text{ W} \tag{43}$$

and

$$\boxed{\sigma_{\text{Conv}}^2 = 1.640750 \times 10^{-10} \text{ W}.} \tag{44}$$

13 13. Fig. 5 Static QuTiP Simulation

The LO-dressed operating point is

$$\frac{\Omega_{\text{LO}}}{2\pi} = 4.23 \text{ MHz}, \tag{45}$$

$$\frac{\Omega_{\text{RF}}}{2\pi} = 20 \text{ kHz}, \tag{46}$$

$$\Delta_f = 150 \text{ kHz}, \tag{47}$$

$$\Delta_c = 0. \tag{48}$$

The static quantum response is generated by sweeping

$$0 \leq \frac{\Omega_{\text{total}}}{2\pi} \leq 16 \text{ MHz} \tag{49}$$

with

$$801 \tag{50}$$

QuTiP points in the Fig. 5 static-response implementation.
The response is

$$\Omega_{\text{total}} \longrightarrow \rho_{21} \longrightarrow \chi \longrightarrow P_{\text{out}}. \tag{51}$$

14 14. Linear Dynamic Range and κ

The linear dynamic range is determined using an operating-point-anchored linear fit with

$$R^2 \geq 0.995. \tag{52}$$

The wider Fig. 6 QuTiP sweep uses

$$0 \leq \Omega_{\text{total}}/(2\pi) \leq 25 \text{ MHz}, \tag{53}$$

$$N_{\text{sweep}} = 641. \tag{54}$$

The resulting linear region is approximately

$$\boxed{0.4688 \leq \frac{\Omega_{\text{total}}}{2\pi} \leq 7.9688 \text{ MHz}.} \tag{55}$$

The fitted slope is

$$\boxed{\kappa = -2.269728 \times 10^{-7} \text{ W/MHz}.} \tag{56}$$

In Eq. (37) angular-frequency units,

$$\boxed{\kappa = -3.612384 \times 10^{-14} \text{ W/(rad/s)}.} \tag{57}$$

The real QuTiP output at the LO operating point is

$$\boxed{\overline{P}_0 = P_{\text{out}}(\Omega_{\text{LO}}) \approx 8.8508 \ \mu\text{W}.} \tag{58}$$

15 15. LO-Dressed Noise Quantities

The coefficient multiplying P_{Rx} in the LO-dressed signal term is

$$A_{\text{LO}} = G_{\text{LNA}} R_L D^2 \kappa^2 \left(\frac{\wp_{\text{RF}}}{\hbar}\right)^2. \tag{59}$$

The current implementation gives

$$\boxed{A_{\text{LO}} = 2.559684 \times 10^{-8}.} \tag{60}$$

The thermal-noise contribution is

$$\sigma_{\text{TN}}^2 = 4k_B T B = 1.601553 \times 10^{-14} \text{ W}. \tag{61}$$

The photon-shot-noise contribution is

$$\boxed{\sigma_{\text{PSN}}^2 = 1.554911 \times 10^{-18} \text{ W}.} \tag{62}$$

The BGN-driven contribution is

$$A_{\text{LO}} \sigma_{\text{BGN}}^2 = 2.559684 \times 10^{-20} \text{ W}. \tag{63}$$

After multiplication by the photodiode/LNA/load factors,

$$G_{\text{LNA}} R_L D^2 = 1512.5. \tag{64}$$

Therefore,

$$\text{BGN contribution} = 2.559684 \times 10^{-20}\ \text{W}, \tag{65}$$
$$\text{PSN contribution} = 2.351286 \times 10^{-15}\ \text{W}, \tag{66}$$
$$\text{thermal contribution} = 1.601553 \times 10^{-14}\ \text{W}. \tag{67}$$

The total LO-dressed noise variance is

$$\boxed{\sigma^2_{\text{Ry,LO}} = 1.836736 \times 10^{-14}\ \text{W}.} \tag{68}$$

16 16. Fig. 6 Checkpoint

For

$$P_{\text{Tx}} = -10\ \text{dBm} \tag{69}$$

and

$$d_{\text{Tx}-\text{Rx}} = 100\ \text{m}, \tag{70}$$

the transmitted power is

$$P_{\text{Tx}} = 10^{-4}\ \text{W}. \tag{71}$$

The conventional received power is

$$\boxed{P_c(100\ \text{m}) = 7.622292 \times 10^{-13}\ \text{W}.} \tag{72}$$

The conventional SNR is

$$\boxed{\text{SNR}_{\text{Conv}} = -23.329568\ \text{dB}.} \tag{73}$$

The theoretical LO-dressed signal term is

$$1.259847 \times 10^{-14}. \tag{74}$$

Hence,

$$\boxed{\text{SNR}_{\text{LO,theory}} = -1.637288\ \text{dB}.} \tag{75}$$

The difference is

$$\boxed{\Delta\text{SNR} = 21.692280\ \text{dB}.} \tag{76}$$

The paper claims approximately 44 dB for this comparison. Therefore, the current independent reproduction does not reproduce the paper's claimed absolute SNR advantage.

17 17. Fig. 4 LO-Free AT-Splitting Simulation

The LO-free atomic characterization sweeps

$$\frac{\Omega_{\rm RF}}{2\pi} = 1\ \text{MHz} \quad \text{to} \quad 10\ \text{MHz} \tag{77}$$

using

$$181 \tag{78}$$

RF points in the characterization procedure.
For each RF Rabi frequency, the coupling detuning is swept over

$$-10\ \text{MHz} \leq \frac{\Delta_c}{2\pi} \leq 10\ \text{MHz} \tag{79}$$

using

$$2001 \tag{80}$$

points.
The theoretical AT splitting is

$$\boxed{\Delta\nu_{\rm AT} \approx \frac{|\Omega_{\rm RF}|}{2\pi}.} \tag{81}$$

The current validated threshold is

$$\boxed{\frac{\Omega_{\rm RF,th}}{2\pi} = 5.500\ \text{MHz}.} \tag{82}$$

18 18. Fig. 6 Distance / Transmit-Power Sweeps

The current implementation uses the following plotted transmit-power sets:

Table 7: Transmit powers used by the Fig. 6 implementation.

Receiver	$P_{\rm Tx}$ values	Status
Conventional	-10, 10 dBm	-10 dBm prose-stated; 10 dBm legend-based
LO-free	10, 20 dBm	implementation from figure legend
LO-dressed	-10, 10 dBm	-10 dBm prose-stated; 10 dBm legend-based

The distance vector is

$$\boxed{d = 10^0, \ldots, 10^6\ \text{m}} \tag{83}$$

with

$$241 \tag{84}$$

logarithmically spaced points.

19 19. Fig. 7 Mutual Information

The conventional curve uses the standard AWGN expression

$$\boxed{I_{\mathrm{Conv}} = \log_2(1 + \mathrm{SNR}).} \tag{85}$$

This formula is an implementation inference from the conventional complex-AWGN channel model; it is not explicitly written as a mutual-information equation in the paper.

The theoretical LO-dressed curve uses the paper's Eq. (38),

$$\boxed{I_{\mathrm{LO}} = e^{1/\mathrm{SNR}_{\mathrm{Ry,LO}}} E_1\left(\frac{1}{\mathrm{SNR}_{\mathrm{Ry,LO}}}\right) \ln 2.} \tag{86}$$

The LO-free mutual information uses the paper's Eq. (22),

$$\boxed{\mathcal{I}(z; x) = \frac{1}{2} \ln\left(\frac{\pi(r+1)}{2}\right) - \ln I_0(r) + r\frac{I_1(r)}{I_0(r)}.} \tag{87}$$

The paper relates

$$r \approx \mathrm{SNR}_{\mathrm{Ry}}. \tag{88}$$

20 20. Fig. 7 Monte-Carlo Settings

The current Monte-Carlo reconstruction used:

Table 8: Monte-Carlo settings used in the Fig. 7 reconstruction.

Quantity	Symbol / setting	Value
SNR range	–	−20 to 40 dB
Number of SNR points	–	81
Trials per point	N	20 000
Random seed	–	42
Input symbols	x_i	$\mathcal{CN}(0, 1)$
Noise	n_i	complex Gaussian
Channel coefficient	h	1
AT threshold	–	5.500 MHz

The Monte-Carlo generative model is

$$x_i \sim \mathcal{CN}(0, 1) \tag{89}$$

and

$$n_i \sim \mathcal{CN}(0, \sigma_{\mathrm{Ry}}^2). \tag{90}$$

The per-trial RF Rabi frequency is obtained from the RF-field amplitude through

$$\Omega_{\mathrm{RF}} = \frac{|\wp_{\mathrm{RF}}|}{\hbar}|E_{\mathrm{RF}}|. \tag{91}$$

The current implementation applies the gate

$$\boxed{\frac{\Omega_{\mathrm{RF}}}{2\pi} \geq 5.500 \text{ MHz}} \tag{92}$$

for AT-splitting resolvability.

Important: the exact behaviour of the receiver when the AT splitting is unresolved is not specified by the paper. The noise-only unresolved branch used in the current Monte-Carlo code is therefore an implementation decision, not a paper equation.

21 21. Fig. 7 Important Current Results

The first implementation produced the following checkpoint values:

Table 9: Fig. 7 checkpoint values from the current implementation.

SNR (dB)	Conventional	Theoretical LO	Practical LO	LO-free
−20	0.0144	0.0069	0.0088	unresolved
0	1.0000	0.4134	0.4878	unresolved
20	6.6582	2.8270	2.9939	unresolved
40	13.2879	5.9847	6.1012	unresolved

The LO-free AT threshold maps to approximately

$$\boxed{\mathrm{SNR} \approx 46.02 \text{ dB}} \tag{93}$$

in the current parameterization, which is outside the original −20 to 40 dB Fig. 7 window.

22 22. Main Numerical Warnings to Remember

1. The paper states a probe beam diameter of 0.76 mm, but the frozen Fig. 3 code uses 0.38 mm as an empirical calibration.
2. The resulting calibrated probe input power is $P_{\mathrm{in}} = 9.7690\ \mu\mathrm{W}$.
3. The paper's literal diameter interpretation gives $P_{\mathrm{in}} = 39.0761\ \mu\mathrm{W}$.
4. The current Fig. 6 theoretical LO-dressed SNR at $P_{\mathrm{Tx}} = -10$ dBm and $d = 100$ m is −1.6373 dB.
5. The conventional SNR at the same point is −23.3296 dB.
6. The resulting advantage is 21.6923 dB, not the paper's claimed approximately 44 dB.
7. The fitted κ is not printed numerically in the paper. The current value is obtained from the real QuTiP static response.
8. The current fitted value is
$$\kappa = -3.612384 \times 10^{-14} \text{ W/(rad/s)}.$$
9. The AT-splitting threshold used downstream is 5.500 MHz.
10. The paper does not fully specify how the Practical LO-dressed and LO-free Fig. 7 curves were generated from the phrase "QuTiP and Monte-Carlo simulations."
11. The current Monte-Carlo unresolved-AT branch is therefore an implementation choice and must not be described as an equation explicitly supplied by the paper.

23. One-Line Physical Picture

The entire numerical chain can be remembered as

$$\boxed{P_{\mathrm{Tx}} \to P_{\mathrm{Rx}} \to E_{\mathrm{RF}} \to \Omega_{\mathrm{RF}} \to H \to \rho_{\mathrm{ss}} \to \rho_{21} \to \chi \to P_{\mathrm{out}} \to \kappa, \text{ noise} \to \mathrm{SNR} \to \text{Mutual Information.}} \tag{94}$$

For the quantum part:

$$\boxed{\text{QuTiP atomic characterization} \to \text{response lookup table} \to \text{communication / Monte-Carlo simulation.}} \tag{95}$$

End of simulation parameter sheet